

1)pwd

```
[medikondarakesh@Medikondas-MacBook-Air ~ % pwd
/Users/medikondarakesh
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

2)Clear

```
[medikondarakesh@Medikondas-MacBook-Air ~ % pwd
/Users/medikondarakesh
[medikondarakesh@Medikondas-MacBook-Air ~ % clear
```

3)history

```
[medikondarakesh@Medikondas-MacBook-Air ~ % history
 987  pwd
 988  $
 989  pwd
 990  clear
 991  pwd
 992  clear
 993  pwd
 994  clear
 995  pwd
 996  clear
 997  pwd
 998  clear
 999  pwd
1000  clear
1001  pwd
1002  clear
```

4)whoami

```
[medikondarakesh@Medikondas-MacBook-Air ~ % history
 987  pwd
 988  $
 989  pwd
 990  clear
 991  pwd
 992  clear
 993  pwd
 994  clear
 995  pwd
 996  clear
 997  pwd
 998  clear
 999  pwd
1000  clear
1001  pwd
1002  clear
[medikondarakesh@Medikondas-MacBook-Air ~ % whoami
medikondarakesh
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

5)date

```
1002  clear
[medikondarakesh@Medikondas-MacBook-Air ~ % whoami
medikondarakesh
[medikondarakesh@Medikondas-MacBook-Air ~ % date
Sat Aug  8 19:40:05 IST 2026
█
```

6)cal

```
30 31
[medikondarakesh@Medikondas-MacBook-Air ~ % cal
  August 2026
Su Mo Tu We Th Fr Sa
      1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
```

7)hostname

```
[medikondarakesh@Medikondas-MacBook-Air ~ % hostname
Medikondas-MacBook-Air.local
```

8)uname -a

```
[medikondarakesh@Medikondas-MacBook-Air ~ % uname -a
Darwin Medikondas-MacBook-Air.local 25.5.0: Tue Jun  9 22:26:22 PDT 2026; root:xnu-12377.121.10~1/RELEASE_ARM64_T8132 arm64
[medikondarakesh@Medikondas-MacBook-Air ~ %
```

9)exit

```
[medikondarakesh@Medikondas-MacBook-Air ~ % exit
```

```
Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.
```

```
[Process completed]
```

```

medikondarakesh@Medikondas-MacBook-Air ~ % ls
2520030001_Practical    BloodBankSystem.java    Downloads                Movies                    Red.c.save
2520030001_Skill        Buffer.java              For.class                Music                    report.txt
2520030002_Practical    Buffered.java           For.java                 my-app                  Riot.txt
2520030002_Skill        BufferedReader.java      ForgeOS                  npm                      Ruok.java
2520030003_Practical    Calculator.class         hello                    Numbers.class           s.circ
2520030003_Skill        CardPayment.class        hello.c                  Numbers.java            ShoppingCart.class
2520030035_OSSP         Case.class               hello.c                  Os_rak                  TestStrategy.java
2520030448_OSSP         Case.java                hello.c gcc hello.c -o hello    OS cd                   Time.class
Applications            Case.java                hello.java               OS pwd                  Time.java
Ar.java                 ddcalab.circ             import java.util.java     paymentStrategy.class   TwoDimension.java
ark.txt                 Desktop                  library                   Pictures                 Upipayment.class
ArrayEx.java            DickJ.java               logisim evolution        Public                   username
arrays.java             Documents                Main.class               R.html                  username.c
BloodBankSystem.class   Donor.class              Main.java                Rakesh                  WalletPayment.class
medikondarakesh@Medikondas-MacBook-Air ~ %

```

10)ls

11)ls -l

```

-rw-r--r--@ 1 medikondarakesh staff 240 19 Nov 2025 Ruok.java
-rw-r--r--@ 1 medikondarakesh staff 7893 1 Dec 2025 s.circ
-rw-r--r--@ 1 medikondarakesh staff 448 1 Dec 2025 ShoppingCart.class
-rw-r--r--@ 1 medikondarakesh staff 724 1 Dec 2025 TestStrategy.java
-rw-r--r--@ 1 medikondarakesh staff 1052 24 Nov 2025 Time.class
-rw-r--r--@ 1 medikondarakesh staff 268 24 Nov 2025 Time.java
-rw-r--r--@ 1 medikondarakesh staff 281 19 Nov 2025 TwoDimension.java
-rw-r--r--@ 1 medikondarakesh staff 863 1 Dec 2025 Upipayment.class
-rwxr-xr-x 1 medikondarakesh staff 33592 8 Aug 08:31 username
-rw-r--r-- 1 medikondarakesh staff 134 8 Aug 08:31 username.c
-rw-r--r--@ 1 medikondarakesh staff 868 1 Dec 2025 WalletPayment.class
medikondarakesh@Medikondas-MacBook-Air ~ %

```

12)ls -a

```
medikondarakesh@Medikondas-MacBook-Air ~ % ls -a
.                .Trash
..               .viminfo
.arduinoIDE      .vscode
.cache           .vscode-shared
.CFUserTextEncoding .zsh_history
.codex           .zsh_sessions
.copilot         2520030001_Practical
.ddcalab.circ.autosave 2520030001_Skill
.DS_Store        2520030002_Practical
.gemini          2520030002_Skill
.gitconfig       2520030003_Practical
.hushlogin       2520030003_Skill
.lessshst        2520030035_OSSP
.lima            2520030448_OSSP
.local           Applications
.matplotlib      Ar.java
.npm             ark.txt
.npmrc           ArrayEx.java
.ssh             arrays.java
.swiftpm         BloodBankSystem.class
medikondarakesh@Medikondas-MacBook-Air ~ %
```

13)mkdir Documents

```
medikondarakesh@Medikondas-MacBook-Air ~ % mkdir Documents
mkdir: Documents: File exists
medikondarakesh@Medikondas-MacBook-Air ~ % cd
medikondarakesh@Medikondas-MacBook-Air ~ %
```

14) rmdir

```
ondarakesh@Medikondas-MacBook-Air ~ % cd
ondarakesh@Medikondas-MacBook-Air ~ % mkdir Test && rmdir Test
ondarakesh@Medikondas-MacBook-Air ~ %
```

15) find

```
medikondarakesh@Medikondas-MacBook-Air ~ % touch hello.c
medikondarakesh@Medikondas-MacBook-Air ~ % mdfind -name hello.c
2026-08-08 20:04:27.680 mdfind[28782:3619646] [UserQueryParser] Loading keywords and predicates for locale "en_US"
2026-08-08 20:04:27.680 mdfind[28782:3619646] [UserQueryParser] Loading keywords and predicates for locale "en"
/Users/medikondarakesh/hello.c
/Users/medikondarakesh/hello.c
/Users/medikondarakesh/hello.c gcc hello.c -o hello
/Users/medikondarakesh/2520030448_OSSP/Practical/hello.c
/Users/medikondarakesh/2520030035_OSSP/Practical/hello.c
/Users/medikondarakesh/2520030035_OSSP/hello.c
medikondarakesh@Medikondas-MacBook-Air ~ %
```

16) locate

```
medikondarakesh@Medikondas-MacBook-Air ~ % mdfind -name hello.c
2026-08-08 20:04:27.680 mdfind[28782:3619646] [UserQueryParser] Loading keywords and predicates for locale "en_US"
2026-08-08 20:04:27.680 mdfind[28782:3619646] [UserQueryParser] Loading keywords and predicates for locale "en"
/Users/medikondarakesh/hello.c
/Users/medikondarakesh/hello.c
/Users/medikondarakesh/hello.c gcc hello.c -o hello
/Users/medikondarakesh/2520030448_OSSP/Practical/hello.c
/Users/medikondarakesh/2520030035_OSSP/Practical/hello.c
/Users/medikondarakesh/2520030035_OSSP/hello.c
medikondarakesh@Medikondas-MacBook-Air ~ % mdfind -name hello.c
2026-08-08 20:07:02.713 mdfind[28812:3622046] [UserQueryParser] Loading keywords and predicates for locale "en_US"
2026-08-08 20:07:02.713 mdfind[28812:3622046] [UserQueryParser] Loading keywords and predicates for locale "en"
/Users/medikondarakesh/hello.c
/Users/medikondarakesh/hello.c
/Users/medikondarakesh/hello.c gcc hello.c -o hello
/Users/medikondarakesh/2520030448_OSSP/Practical/hello.c
/Users/medikondarakesh/2520030035_OSSP/Practical/hello.c
/Users/medikondarakesh/2520030035_OSSP/hello.c
medikondarakesh@Medikondas-MacBook-Air ~ %
```

17) touch

```
medikondarakesh@Medikondas-MacBook-Air ~ % touch file1.txt
medikondarakesh@Medikondas-MacBook-Air ~ %
```

18) cat

```
medikondarakesh@Medikondas-MacBook-Air ~ % touch file1.txt
medikondarakesh@Medikondas-MacBook-Air ~ % echo "Hello World" > file1.txt && cat file1.txt
Hello World
medikondarakesh@Medikondas-MacBook-Air ~ %
```

19)head

```
medikondarakesh@Medikondas-MacBook-Air ~ % head hello.c
```

```
#include <stdio.h>
int main(){
    printf("Hello Operating Systems!\n");
    return 0;
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

20)tail

```
medikondarakesh@Medikondas-MacBook-Air ~ % tail hello.c
```

```
#include <stdio.h>
int main(){
    printf("Hello Operating Systems!\n");
    return 0;
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

21)more

```
medikondarakesh@Medikondas-MacBook-Air ~ % more file1.txt
Hello World
file1.txt (END)
```

22)less

```
#include <stdio.h>
int main(){
    printf("Hello Operating Systems!\n");
    return 0;
}
hello.c (END)
```

23)wc

```
medikondarakesh@Medikondas-MacBook-Air ~ % wc hello.c
   7      10      91 hello.c
medikondarakesh@Medikondas-MacBook-Air ~ %
```

24)diff

25)sort

```
medikondarakesh@Medikondas-MacBook-Air ~ % printf "Zebra\nApple\nMango\n" > data.txt && sort data.txt
Apple
Mango
Zebra
medikondarakesh@Medikondas-MacBook-Air ~ %
```

26)uniq

```
medikondarakesh@Medikondas-MacBook-Air ~ % printf "Apple\nApple\nMango\n" > data.txt && uniq data.txt # Filter out repeated adjacent lines
usage: uniq [-cdiu] [-D[septype]] [-f fields] [-s chars] [input [output]]
medikondarakesh@Medikondas-MacBook-Air ~ %
```

27)grep

```
medikondarakesh@Medikondas-MacBook-Air ~ % grep main hello.c
int main(){
medikondarakesh@Medikondas-MacBook-Air ~ %
```

28)nan0

```
UW PICO 5.09 File: hello.c
#include <stdio.h>
int main(){
    printf("Hello Operating Systems!\n");
    return 0;
}
```

30)vi

```
#include <stdio.h>
int main(){
    printf("Hello Operating Systems!\n");
    return 0;
}
```

31)gcc

```
Restored session: Sat Aug 8 22:24:22 IST 2026
medikondarakesh@Medikondas-MacBook-Air ~ % gcc hello.c -o hello
medikondarakesh@Medikondas-MacBook-Air ~ %
```

32)./program

```
medikondarakesh@Medikondas-MacBook-Air ~ % gcc hello.c -o hello
medikondarakesh@Medikondas-MacBook-Air ~ % ./hello
Hello Operating Systems!
medikondarakesh@Medikondas-MacBook-Air ~ %
```

33)which

```
medikondarakesh@Medikondas-MacBook-Air ~ % which gcc
/usr/bin/gcc
medikondarakesh@Medikondas-MacBook-Air ~ %
```

34)where

```
/usr/bin/gcc
medikondarakesh@Medikondas-MacBook-Air ~ % whereis gcc
gcc: /usr/bin/gcc
medikondarakesh@Medikondas-MacBook-Air ~ %
```

35)ps

```
medikondarakesh@Medikondas-MacBook-Air ~ % ps
  PID TTY          TIME CMD
 29467 ttys000      0:00.01 -zsh
 29470 ttys001      0:00.03 -zsh
medikondarakesh@Medikondas-MacBook-Air ~ %
```

36)ps -ef

```
501 29954      1  0 10:03AM ??          0:34.96 /System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.WebContent.xpc.
WebContent
501 29960      1  0 10:08AM ??          0:00.28 /System/Library/PrivateFrameworks/ShareKit.framework/Versions/A/PlugIns/ShareSheetUI.appex/Content:
ges ("en-US")
501 29961      1  0 10:09AM ??          0:00.06 /System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.WebContent.xpc.
WebContent
0 29466 29465      0  8:29AM ttys000  0:00.01 login -pf medikondarakesh
501 29467 29466      0  8:29AM ttys000  0:00.01 -zsh
0 29468 29465      0  8:29AM ttys001  0:00.01 login -pf medikondarakesh
```

37)top

```
com.apple.appkit.xpc.openAndSavePanelService
501 29372 1 0 7:25AM ?? 0:00.25 /System/Library/Frameworks/QuickLookUI.framework/Versions/A/XPCServices/QuickLookUISer
501 29373 1 0 7:25AM ?? 0:00.01 /System/Library/Frameworks/NetFS.framework/Versions/A/XPCServices/PlugInLibraryService
501 29374 1 0 7:25AM ?? 0:00.26 /System/Library/Frameworks/AppKit.framework/Versions/C/XPCServices/ThemeWidgetControlV
rolViewService
501 29463 1 0 8:28AM ?? 0:06.74 /System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.We
WebContent
501 29464 1 0 8:28AM ?? 0:15.42 /System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.We
WebContent
501 29465 1 0 8:29AM ?? 0:10.66 /System/Applications/Utilities/Terminal.app/Contents/MacOS/Terminal
501 29500 1 0 8:32AM ?? 1:41.57 /System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.We
WebContent
501 29655 1 0 8:59AM ?? 0:00.28 /System/Library/PrivateFrameworks/CoreParsec.framework/parsec-fbf
501 29658 1 0 8:59AM ?? 0:00.66 /usr/libexec/remindd
501 29834 1 0 9:14AM ?? 0:00.34 /System/iOSSupport/System/Library/PrivateFrameworks/HomeEnergyDaemon.framework/Support
501 29903 1 0 9:28AM ?? 0:00.25 /System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.We
WebContent
501 29916 1 0 9:36AM ?? 0:03.32 /usr/libexec/spotlightknowledge.updater -u
501 29954 1 0 10:03AM ?? 0:34.96 /System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.We
WebContent
501 29960 1 0 10:08AM ?? 0:00.28 /System/Library/PrivateFrameworks/ShareKit.framework/Versions/A/PlugIns/ShareSheetUI.a
ges ("en-US")
501 29961 1 0 10:09AM ?? 0:00.06 /System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.We
WebContent
0 29466 29465 0 8:29AM ttys000 0:00.01 login -pf medikondarakesh
501 29467 29466 0 8:29AM ttys000 0:00.01 -zsh
0 29468 29465 0 8:29AM ttys001 0:00.01 login -pf medikondarakesh
501 29470 29468 0 8:29AM ttys001 0:00.04 -zsh
0 29968 29470 0 10:13AM ttys001 0:00.01 ps -ef
medikondarakesh@Medikondas-MacBook-Air ~ % top[cite: 1]
zsh: bad pattern: top[cite:
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

38)kill

```
medikondarakesh@Medikondas-MacBook-Air ~ %
[Restored 9 Aug 2026 at 10:28:08 AM]
medikondarakesh@Medikondas-MacBook-Air ~ % kill $!
medikondarakesh@Medikondas-MacBook-Air ~ % kill -9 $!
```

[Process completed]█

39)jobs

```
medikondarakesh@Medikondas-MacBook-Air ~ % sleep 1000 &
[1] 30048
medikondarakesh@Medikondas-MacBook-Air ~ % jobs
[1] + running    sleep 1000
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

40)fg and bg

```
medikondarakesh@Medikondas-MacBook-Air ~ % fg %1
[1] + running    sleep 1000
^Z
zsh: suspended  sleep 1000
medikondarakesh@Medikondas-MacBook-Air ~ % bg %1
[1] + continued  sleep 1000
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

41)&

```
[1] + continued  sleep 1000
medikondarakesh@Medikondas-MacBook-Air ~ % ./hello $
Hello Operating Systems!
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

42)chmod

```
medikondarakesh@Medikondas-MacBook-Air ~ % touch script.sh
medikondarakesh@Medikondas-MacBook-Air ~ % chmod +x script.sh
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

43)Chmod 755

```
medikondarakesh@Medikondas-MacBook-Air ~ % chmod 755 script.sh
medikondarakesh@Medikondas-MacBook-Air ~ %
```

44)ln

```
medikondarakesh@Medikondas-MacBook-Air ~ % ln file1.txt file2.txt
medikondarakesh@Medikondas-MacBook-Air ~ %
```

45)ln -s

```
medikondarakesh@Medikondas-MacBook-Air ~ % ln file1.txt file2.txt
medikondarakesh@Medikondas-MacBook-Air ~ % ln -s file1.txt shortcut
medikondarakesh@Medikondas-MacBook-Air ~ %
```

46)df -h

```
medikondarakesh@Medikondas-MacBook-Air ~ % ln file1.txt file2.txt
medikondarakesh@Medikondas-MacBook-Air ~ % ln -s file1.txt shortcut
medikondarakesh@Medikondas-MacBook-Air ~ % df -h
```

Filesystem	Size	Used	Avail	Capacity	used	ifree	%iused	Mounted on
/dev/disk2s3s1	228Gi	12Gi	76Gi	14%	459k	795M	0%	/
devfs	215Ki	215Ki	0Bi	100%	744	0	100%	/dev
/dev/disk2s6	228Gi	24Ki	76Gi	1%	0	795M	0%	/System/Volumes/VM
/dev/disk2s4	228Gi	8.4Gi	76Gi	10%	1.4k	795M	0%	/System/Volumes/Preboot
/dev/disk2s2	228Gi	46Mi	76Gi	1%	110	795M	0%	/System/Volumes/Update
/dev/disk1s2	500Mi	6.0Mi	479Mi	2%	1	4.9M	0%	/System/Volumes/xarts
/dev/disk1s1	500Mi	6.0Mi	479Mi	2%	32	4.9M	0%	/System/Volumes/iSCPreboot
/dev/disk1s3	500Mi	4.6Mi	479Mi	1%	96	4.9M	0%	/System/Volumes/Hardware
/dev/disk2s1	228Gi	131Gi	76Gi	64%	1.5M	795M	0%	/System/Volumes/Data
map auto_home	0Bi	0Bi	0Bi	100%	0	0	-	/System/Volumes/Data/home
/dev/disk4s1	1.4Gi	1.4Gi	2.4Mi	100%	11k	4.3G	0%	/Volumes/ChatGPT Installer
/dev/disk5s1	596Mi	539Mi	57Mi	91%	7.9k	4.3G	0%	/Volumes/Arduino IDE 2.3.10
/dev/disk6	1.2Gi	1.2Gi	8.3Mi	100%	28k	4.3G	0%	/Volumes/MySQL Workbench community-8.0.47
/dev/disk7s1	600Mi	560Mi	40Mi	94%	4	4.3G	0%	/Volumes/mysql-9.7.1-macos15-arm64
/dev/disk8s1	951Mi	846Mi	105Mi	89%	8.2k	4.3G	0%	/Volumes/VS Code

```
medikondarakesh@Medikondas-MacBook-Air ~ %
```

47)du -sh

```
du: ./Downloads: Operation not permitted
30G
medikondarakesh@Medikondas-MacBook-Air ~ %
```

49)free

```
medikondarakesh@Medikondas-MacBook-Air ~ % top -l 1 | grep PhysMem
PhysMem: 15G used (1994M wired, 2301M compressor), 176M unused.
medikondarakesh@Medikondas-MacBook-Air ~ %
```

50)id

```
medikondarakesh@Medikondas-MacBook-Air ~ % id
uid=501(medikondarakesh) gid=20(staff) groups=20(staff),12(everyone),61(localaccounts),79(_appserverusr),80(admin),81(_appserveradm),701(com.apple.sharepoint.group.1),33(_appstore),98(_lpadmin),100(_lpoperator),204(_developer),250(_analyticsusers),395(com.apple.access_ftp),398(com.apple.access_screensharing),399(com.apple.access_ssh),400(com.apple.access_remote_ee)
medikondarakesh@Medikondas-MacBook-Air ~ %
```

51)env

```
medikondarakesh@Medikondas-MacBook-Air ~ % env
__CFBundleIdentifier=com.apple.Terminal
TMPDIR=/var/folders/n8/_lnmfc4n07b64h3t2rh_kx4w0000gn/T/
XPC_FLAGS=0x0
TERM=xterm-256color
SSH_AUTH_SOCK=/var/run/com.apple.launchd.K66LjvJ0z9/Listeners
XPC_SERVICE_NAME=0
TERM_PROGRAM=Apple_Terminal
COLORTERM=truecolor
OSLogRateLimit=64
medikondarakesh@Medikondas-MacBook-Air ~ % export Path=$Path:.
medikondarakesh@Medikondas-MacBook-Air ~ %
```

```
y.cryptexd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin:/Library/Apple/usr/bin
SHLVL=1
PWD=/Users/medikondarakesh
OLDPWD=/Users/medikondarakesh
LANG=en_US.UTF-8
_=/usr/bin/env
medikondarakesh@Medikondas-MacBook-Air ~ %
```

52)export

53)alias

```
[medikondarakesh@Medikondas-MacBook-Air ~ % alias ll='ls-l'
medikondarakesh@Medikondas-MacBook-Air ~ % █
```